

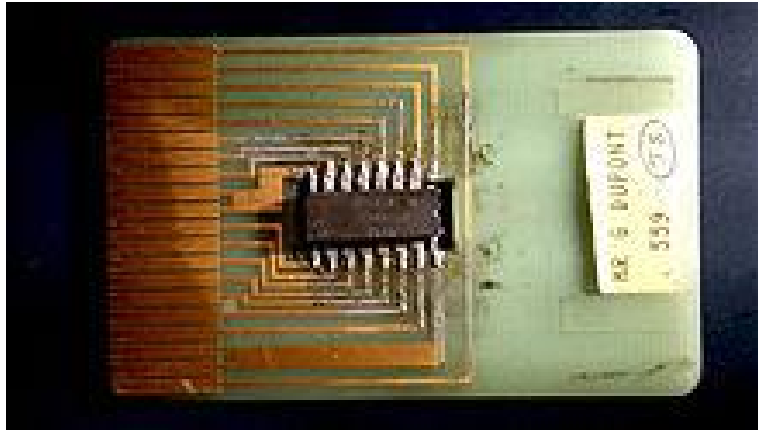
- Transaction Cards:
 - Introduction to Transaction Cards
 - ◆ What are Transaction Cards?

Transaction Cards (Payment cards) are part of a payment system issued by financial institutions, such as a bank, to a customer that enables its owner (the cardholder) to access the funds in the customer's designated bank accounts, or through a credit account and make payments by electronic transfer with a payment terminal and access [automated teller machines \(ATMs\)](#). Such cards are known by a variety of names, including **bank cards**, **ATM cards**, **client cards**, **key cards** or **cash cards**.

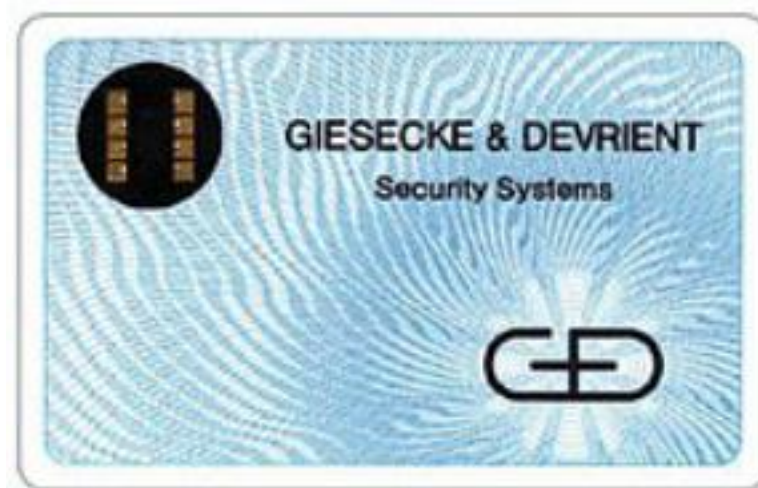
- ◆ History and Evolution of Transaction Cards
 - [Cheque Guarantee Card](#)
 - A cheque guarantee card was an abbreviated portable letter of credit granted by a bank to a qualified depositor in the form of a plastic card that was used in conjunction with a cheque.
 - Eurocheque guarantee card issued by Dortmunder Volksbank till 1993



- The first bank cards were **automated teller machine (ATM) cards** issued by **Barclays** in **London** in **1967**,
- and by **Chemical Bank** in **Long Island, New York**, in **1969**.
- In **1972**, **Lloyds Bank** issued the first bank card to feature an **information-encoding magnetic strip**, using a **personal identification number (PIN)** for security.
- The **smart card**, which was introduced in the **1970s**, was adapted for use as a payment card. Smart payment cards were introduced to the banking sector in **1979**, and came into wider use in the **mid-1980s**.



One of the first smart card prototypes, created by its inventor Roland Moreno around 1975. The chip has not yet been miniaturized. On this prototype, one can see how each pin of the microchip (center) is connected to the exterior world by a copper connector.



First smart card manufactured by Giesecke & Devrient in 1979, already with the finally standardized dimension (ID-1) and a contact area with eight pads (initially on the upper left corner)

- **Types of Cards**

There are a number of types of payment cards, the most common being credit cards, debit cards, charge cards, and prepaid cards.

Most commonly, a payment card is electronically linked to an account or accounts belonging to the cardholder. These accounts may be deposit accounts or loan or credit accounts, and the card is a means of authenticating the cardholder. The largest global card payment organizations are: UnionPay, Visa, Mastercard and American Express.

- **Credit Cards**

- a credit card is linked to a line of credit (usually called a credit limit) created by the issuer of the credit card for the cardholder on which the

cardholder can draw (i.e. borrow), either for payment to a merchant for a purchase or as a cash advance to the cardholder.

- Interest is charged on the portion of the balance not paid off by the due date.



■ Debit Cards

- With a debit card (also known as a bank card, check card or plastic card) when a cardholder makes a purchase, funds are withdrawn directly either from the cardholder's bank account, or from the remaining balance on the card, instead of the holder repaying the money at a later date.

- The use of debit cards has become widespread in many countries and has overtaken use of cheques

- Debit cards can also allow instant withdrawal of cash, acting as the ATM card, and as a cheque guarantee card.

- Merchants usually do not charge a fee for purchases by debit card.

■ Prepaid Cards

A prepaid card, also known as a stored value card, is a payment card that holds a pre-loaded amount of money, allowing users to make purchases without needing a bank account or credit.

The term stored-value card means that the funds and or data are physically stored on the card. With prepaid cards the data is maintained on computers controlled by the card issuer. The value stored on the card can be accessed using a magnetic stripe embedded in the card, on which the card number is encoded; using radio-frequency identification (RFID); or by entering a code number, printed on the card.

■ Virtual credit Cards

What is a virtual credit card?

Virtual credit cards are digital credit cards that function like physical credit cards—but without the plastic. Rather than providing the same card number to multiple vendors, companies can limit their exposure to potential fraud by generating different 16-digit card numbers and expiration dates for each transaction. With the unique number, the payee can only access the approved amount of funds.

- Virtual business card programs can instantly generate unique 16-digit card numbers and expiration dates for a single transaction, so a specific payee can only access the approved amount of funds.
- Virtual cards offer multiple benefits, including greater visibility, security and controls, plus improved expense management.
- When looking for a virtual card provider, it's important to consider integration with existing systems and employee training.

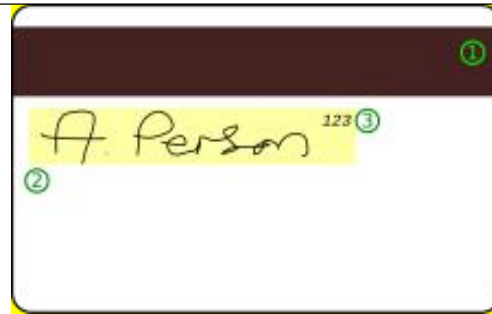
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● Elements in Transaction Cards



An example of the front of a typical debit card:

1. Issuing bank logo
2. EMV chip (optional and may depend on the issuing institution or bank)
3. Hologram (this is located on the back on some cards, including most MasterCards)
4. Card number (PAN) (may vary in length but mostly 16-digits with unique last 4 digits. However in cases such as Discover, Diner's Club, UnionPay & American Express it has a unique 15-digit card number)
5. Card brand logo
6. Expiration date
7. Cardholder's name



An example of the reverse side of a typical debit card:

1. Magnetic stripe
2. Signature strip panel
3. Card Security Code

■ Card Number



Partial IIN on a credit card (both printed and embossed)

IINs and PANs have a certain level of internal structure and share a common numbering scheme set by ISO/IEC 7812. The parts of the number are as follows:

- a six or eight-digit Issuer Identification Number (IIN), the first digit of which is the **major industry identifier** (MII)
- a variable length (up to 12 digits) individual account identifier
- a single **check digit** calculated using the **Luhn algorithm**

1. IIN length has been extended to 10-digits in fifth edition of ISO/IEC 7812 published in 2017 and PAN will continue to remain variable length, ranging from 10 to 19 digits.

Card Number length for different Issuing Network

- **American Express** : 15
- **RuPay**: 16
- **Mastercard**: 16
- **Visa** 13/16/19

■ **Expiry Date** : the format (MM/YY or MM-YY)

■ **CVV Code**

- A **card security code** (CSC; also known as **CVC**, **CVV**, or **several other names**) is a series of numbers that, in addition to the **bank card number**, is printed (but not **embossed**) on a **credit** or **debit card**. The CSC is used as a security feature for **card not present transactions**

-

The card security code is located on the back of Mastercard, Visa, Discover, Diners Club, and JCB credit or debit cards and is typically a separate group of three digits to the right of the signature strip	On American Express cards, the card security code is a printed, not embossed, group of four digits on the front towards the right

Naming

The codes have different names:

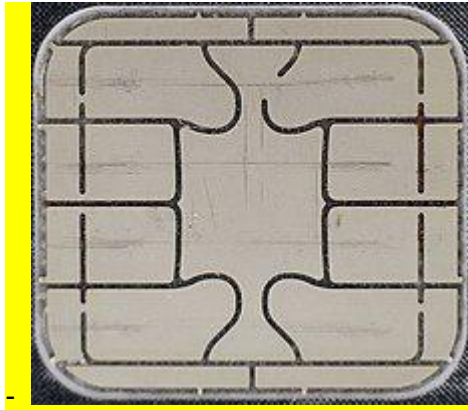
- "CSC" or "card security code": [debit cards](#), American Express (three digits on back of card, also referred to as 3CSC)
- "CVC" or "card validation code": [Mastercard](#)
- "CVV" or "card verification value": [Visa](#)
- "CID": "card ID", "card identification number", or "card identification code": [Discover](#), [American Express](#) (four digits on front of card). American Express usually uses the four-digit code on the front of the card, referred to as the card identification code (CID), but also has a three-digit code on the back of the card, referred to as the card security code (CSC). American Express also sometimes refers to a "unique card code".
- "CVD" or "card verification data": [Discover](#)
- "CVN" or "card validation number", also "card verification number": [China UnionPay](#), [Google Ads](#)
- "CCV" or "card code verification": commonly used in [Canada](#)

Types

There are several types of security codes and PVV (all generated from [DES](#) key in the bank in [HSM](#) modules using [PAN](#), expiration date and service code):

- The first code, 3 numbers, called CVC1 or CVV1, is encoded on track one and two of the [magnetic stripe](#) of the card and used for card present transactions, with signature (second track also contains pin verification value, PVV, but now it is usually all zeroed out and service code). The purpose of the code is to verify that a payment card is actually in the hand of the merchant (thus it should be different from CVV2). This code is automatically retrieved when the magnetic stripe of a card is read (swiped) on a [point-of-sale](#) (card present) device and is verified by the issuer. A limitation is that if the entire card has been duplicated and the magnetic stripe copied, then the code is still valid, even though you usually need to sign after that.
- The second code, and the most cited, is CVV2 or CVC2. This code is often used by merchants for [card not present transactions](#) including online purchases. In some countries in Western Europe, card issuers require a merchant to obtain the code when the cardholder is not present in person. Uses service code 000.
- Contactless and/or chip [EMV](#) cards supply their own electronically generated codes, called iCVV. Uses service code 999. It is described in public standards from EMVCo.
- Consumer Device Cardholder Verification Method (CDCVM for short) is a type of identity verification in which the user's mobile device (such as a smartphone) is used to verify the user's identity; for example, it can use the device's [biometrics](#) authentication features (e.g. [Touch ID](#) or [Face ID](#)), or the device's set [passcode](#). It is supported by a number of payment systems, such as [Apple Pay](#), [Google Pay](#) or [Samsung Pay](#).

■ [EMV Chip](#)



Contact pad for the electrical interface on the front side of a credit card



A smart-card **pinout**.

C1: VCC: Power supply.

C2: RST: Reset signal, used to reset the card's communications.

C3: CLK: Provides the card with a **clock signal**, from which data communications timing is derived.

C5: GND: **Ground** (reference voltage).

C6: VPP: ISO/IEC 7816-3:1997 standard designated this as a programming voltage: an input for a higher voltage to program persistent memory.

C7: I/O: Serial input and output (**half-duplex**).

C4, C8: The two remaining contacts are AUX1 and AUX2 respectively and are used for **USB** interfaces and other uses.

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■ Magnetic Stripe



Front side of the first Magnetic Stripe plastic credit card. Note that the narrow magnetic stripe is on the front of the card. It was later switched to the back side.

- A **magnetic stripe card** is a type of card capable of storing data by **storing it on magnetic material** attached to a plastic card. A computer device can update the card's content. The magnetic stripe is read by swiping it past a **magnetic reading head**. Magnetic stripe cards are commonly used in **credit cards**, **identity cards**,

and transportation tickets. They may also contain a **radio frequency identification (RFID)** tag, a **transponder device** and/or a **microchip** mostly used for **access control** or electronic payment.

There are up to three tracks on magnetic cards known as tracks 1, 2, and 3. Track 3 is virtually unused by the major worldwide networks, and often is not even physically present on the card by virtue of a narrower magnetic stripe. Point-of-sale card readers almost always read track 1, or track 2, and sometimes both, in case one track is unreadable. The minimum cardholder account information needed to complete a transaction is present on both tracks. Track 1 has a higher bit density (210 bits per inch vs. 75), is the only track that may contain alphabetic text, and hence is the only track that contains the cardholder's name.

Track 1 is written with code known as **DEC SIXBIT** plus odd **parity**. The information on track 1 on financial cards is contained in several formats: **A**, which is reserved for proprietary use of the card issuer, **B**, which is described below, **C-M**, which are reserved for use by ANSI Subcommittee X3B10 and **N-Z**, which are available for use by individual card issuers:

Track 1

Format B:

- **Start sentinel** — one character (generally '%')
- **Format code="B"** — one character (alpha only)
- **Primary account number (PAN)** — up to 19 characters. Usually, but not always, matches the **credit card number** printed on the front of the card.
- **Field Separator** — one character (generally '^')
- **Name** — 2 to 26 characters, surnames separated by space if necessary, Surname separator: /
- **Field Separator** — one character (generally '^')
- **Expiration date** — four characters in the form YYMM.
- **Service code** — three characters
- **Discretionary data** — may include Pin Verification Key Indicator (PVKI, 1 character), PIN Verification Value (PVV, 4 characters), **Card Verification Value or Card Verification Code (CVV or CVC, 3 characters)**
- **End sentinel** — one character (generally '?')
- **Longitudinal redundancy check (LRC)** — it is one character and a validity character calculated from other data on the track.

Track 2

This format was developed by the banking industry (ABA). This track is written with a 5-bit scheme (4 data bits + 1 parity), which allows for sixteen possible characters, which are the numbers 0–9, plus the six characters : ; < = > ? . (It may seem odd that these particular punctuation symbols were selected, but by using them the set of sixteen characters matches the **ASCII** range 0x30 through 0x3f.) The data format is as follows:

- **Start sentinel** — one character (generally ';')

- **Primary account number (PAN)** — up to 19 characters. Usually, but not always, matches the **credit card number** printed on the front of the card.
- **Separator** — one character (generally '=')
- **Expiration date** — four characters in the form YYMM.
- **Service code** — three digits. The first digit specifies the interchange rules, the second specifies authorization processing and the third specifies the range of services
- **Discretionary data** — as in track one
- **End sentinel** — one character (generally '?')
- **Longitudinal redundancy check (LRC)** — it is one character and a validity character calculated from other data on the track. Most reader devices do not make the LRC available for display, but use it to verify the input internally to the device.

Service code values common in financial cards:

First digit

- 1: International interchange OK
- 2: International interchange, use **IC (chip)** where feasible
- 5: National interchange only except under bilateral agreement
- 6: National interchange only except under bilateral agreement, use **IC (chip)** where feasible
- 7: No interchange except under bilateral agreement (closed loop)
- 9: Test

Second digit

- 0: Normal
- 2: Contact issuer via online means
- 4: Contact issuer via online means except under bilateral agreement

Third digit

- 0: No restrictions, PIN required
- 1: No restrictions
- 2: Goods and services only (no cash)
- 3: ATM only, PIN required
- 4: Cash only
- 5: Goods and services only (no cash), PIN required
- 6: No restrictions, use PIN where feasible
- 7: Goods and services only (no cash), use PIN where feasible

Programming

Parsing Track 1 and Track 2 can be done with **Regular Expressions**.

Track 1

```
^%B([0-9]{1,19})\^[^\^]{2,26})\^[([0-9]{4})\^[([0-9]{3})\^[([^\?]*)\?}$
```

This **Regex** will capture all of the important fields into the following groups.

- Group 1: Payment card number (PAN)
- Group 2: Name (NM)
- Group 3: Expiration Date (ED)
- Group 4: Service Code (SC)
- Group 5: Discretionary data (DD)

Track 2

$^{\wedge}\backslash;([0-9]\{1,19\})\backslash=([0-9]\{4\}\backslash=)([0-9]\{3\}\backslash=)([^\wedge\?]*)\backslash?\$$

- Group 1: Primary Account Number (PAN)
- =
- Group 2: Expiration date (ED) or =
- Group 3: Service code (SC) or =
- Group 4: Discretionary data (DD) or empty



EMV contactless symbol used on compatible payment terminals. EMV stands for "Europay, Mastercard, and Visa", the three companies that created the standard.

What is a contactless payment?

Contactless payment, or tap-to-pay, is a secure, touch-free payment method for everyday transactions. It uses Near-Field Communication (NFC) and Radio-Frequency Identification (RFID) to establish communication between the consumer's payment device and the merchant's acquiring bank.

The technology has gained immense popularity as it eliminates the need to provide a Credit or Debit Card to the merchant, enter a Personal Identification Number (PIN), or provide a signature. Moreover, there is no need to count cash or hassle about having change. The payment method does not require physical contact between the device and the terminal, making it a hygienic, convenient, and quick option.

The data exchange between a contactless-enabled card and its reader is possible when the range is around 4 cm. This measure prevents the accidental reading of cards and devices. You can check for the Wi-Fi symbol on devices at the Point-of-Sale (POS) terminals of financial institutions, retail stores, and restaurants to understand whether or not they provide contactless payments.

How does contactless payment work?

After understanding the contactless payment meaning, you must learn how it works to assess if it is right for you. Contactless technology uses NFC, which means two devices can communicate when kept close to each other. The step-by-step process of how the contactless payment system works is as follows.

1.

Initialisation : The consumer holds their contactless card, smartphone, or wearable device close to the NFC-enabled Point-of-Sale (POS) terminal.

2.

Communication: The terminal sends out a radio frequency signal detected by the NFC chip in the payment device.

3.

Encryption: The payment information is securely transmitted using encryption to prevent fraud. This often involves tokenisation, where a unique token is generated for each transaction instead of using the card's details.

4.

Authorisation: The payment processor verifies the transaction details and checks for available funds.

5.

Confirmation: The transaction is completed once authorised, and the terminal displays a confirmation message.

The process typically takes only a few seconds, making it significantly faster than traditional card payments.

Advantages of contactless payment

The contactless technology has many benefits for consumers and businesses.

The benefits of contactless payment for businesses are as follows:

1.

Enhanced customer experience

The checkout experience of businesses offering contactless card payments is much smoother and more convenient. This leads to a better customer experience and, in turn, earns their loyalty.

2.

Better operational efficiency

The workforce requirements at a store with tap-to-pay technology are lower. Moreover, the time spent to complete the transaction using contactless payment is less than cash counting or card machine operation. As a result, businesses have shorter lines, increased customer satisfaction, and the ability

to serve more customers in less time, especially during peak hours. This translates to better operational efficiency.

3.

No additional cost

There is no additional processing fee for the contactless service. Merchants and small business owners don't have to pay anything more than they would pay for a regular Credit Card transaction.

4.

Fraud protection

Contactless payment systems are protected from hacking attempts with their security and encryption measures. A robust security system ensures businesses get their payments without any hindrance. Also, most banks protect transactions against fraud.

Challenges and concerns

Contactless payment solutions have a fair share of concerns from the perspective of consumers and businesses.

The disadvantages of contactless payment for businesses are as follows:

1.

Initial setup expenses

The upfront cost is one of the primary challenges for businesses adopting contactless payments. You have to upgrade your POS systems to support the technology and buy new terminals that support NFC. These setup expenses can be significant, especially for small businesses with limited budgets.

2.

Consumer reluctance

Contactless payments have gained immense popularity. However, not all customers are comfortable using them, especially the older generations. They are less familiar with the technology or may distrust it due to security or privacy concerns. This is a challenge for businesses with a diverse customer base. You will have to switch across payment methods - cash, traditional card, and contactless technology - to accommodate the preferences of different customers, which can complicate operations.

3.

Limited transaction amounts

The Reserve Bank of India (RBI) has set a cap of ₹ 5,000 per transaction for contactless payments. While the limit is set to protect against fraud, it can be inconvenient for customers and businesses. If you handle higher-value transactions frequently, you will need to shift to the traditional payment methods for larger amounts, which can slow down the checkout process.

- Entry Modes

- [ATM/PoS](#)

Not all entry methods are available on all cards.



- **Mag Stripe:** Most cards have a magnetic stripe. This is a black stripe on the back of the card.
 - If the card has both a magnetic stripe and an EMV chip and you swipe it through the stripe reader, your device will probably prompt you to insert the card instead.



- **EMV:** Cards that are capable of being inserted will have an EMV chip.
 - This chip can be round or square, and will usually be on the front face of the card. Depending on the design of the card, the chip can be on either the right or the left.
 - The EMV chip is the most secure entry method.



- **Contactless:** Cards that support "touch-free" or "tap-to-pay" payments will display the Contactless symbol. Your payment information is transferred when you place your card within a few inches of a contactless-enabled payment terminal.
 - Contactless technology is also what is used when you use a payment-enabled mobile or wearable device.
- For debit transactions, the payment card must be swiped, tapped, or inserted. The payment card should not be manually typed into the payment device.

- [Accounts](#)

Types of Banking Accounts

1. Current Account

A current account is designed primarily for business owners, traders, and entrepreneurs who need to make frequent transactions.

Key Features:

- **Overdraft Facility:** Allows you to withdraw more money than what is available in your account.
- **Free Transactions:** Free daily transactions, making it ideal for businesses with high transaction volumes.
- **No Interest Earned:** Unlike savings accounts, current accounts do not earn interest on deposits.
- **Minimum Balance Requirement:** Typically requires a higher minimum balance than other account types.
- **Ideal For:** Businesses and professionals who need to manage a large number of daily transactions.

2. Savings Account

A **savings account** is a standard deposit account where you can save money while earning interest on your deposits.

Key Features:

- **Interest Earnings:** Earns interest on the deposited amount, though the rate may vary between banks.
- **Limited Transactions:** Usually limited to a certain number of transactions per month.
- **Types of Savings Accounts:** Regular Savings Accounts, Children's Savings Accounts, Senior Citizens' Savings Accounts, Women's Savings Accounts, Institutional and Family Savings Accounts
- **Additional Features:** Zero-balance accounts, auto sweep facilities, debit cards, bill payments, and cross-product benefits like discounts on demat accounts.
- **Ideal For:** Individuals looking to save money and earn interest, with options tailored for different demographics and needs.

3. Salary Account

A salary account is a **type of savings account** opened by employers for their employees to facilitate the direct credit of salaries.

Key Features:

- **No Minimum Balance:** Typically, no minimum balance requirement as long as the salary is credited monthly.
- **Additional Benefits:** Often comes with added benefits like higher transaction limits, special offers, and lower loan interest rates.
- **Reimbursement Accounts:** Separate accounts for allowances and reimbursements.

- **Ideal For:** Employees who want a hassle-free way to receive their salary with additional banking benefits.

4. Fixed Deposit Account

A **fixed deposit** (FD) account allows you to earn a higher interest rate on a lump sum amount deposited for a fixed period.

Key Features:

- **Fixed Interest Rate:** Earns a fixed interest rate over the deposit tenure, which can range from seven days to ten years.
- **Premature Withdrawal:** Generally not allowed without a penalty, although some banks offer this feature with lower interest rates.
- **Safe Investment:** Provides a secure way to earn a guaranteed return on your savings.
- **Ideal For:** Individuals looking to invest a lump sum for a guaranteed return over a fixed period.

5. Recurring Deposit Account

A recurring deposit (RD) account allows you to invest a fixed amount regularly and earn interest on it.

Key Features:

- **Regular Investments:** Requires regular deposits (monthly or quarterly) for a fixed tenure.
- **Fixed Tenure:** The tenure cannot be changed once set, and premature withdrawals attract a penalty.
- **Interest Earnings:** Interest is earned on the cumulative deposits and is typically compounded quarterly.
- **Ideal For:** Individuals who want to save regularly and earn interest over a fixed period without investing a lump sum upfront.

6. DEMAT Account

A Demat Account is a digital account that holds shares, bonds and other securities. It offers ease of trading because actual physical certificates (showing ownership of the securities) are not required and transactions are quicker, safer and more convenient. Investors use it to buy, sell and transfer securities easily. In India, depository institutions like NSDL and CDSL oversee these accounts, which are necessary for stock market trading. Demat Accounts have lower risk of theft and loss since they store physical securities in an electronic form.

7. NRI Accounts

NRI accounts are designed for Indians living abroad, offering various options to manage their earnings and savings in India.

Types of NRI Accounts:

1. Non-Resident Ordinary (NRO) Account:

- **Rupee Account:** Allows NRIs to manage income earned in India.
- **Taxable Interest:** Interest earned is subject to tax in India.
- **Convertibility:** Funds can be repatriated after paying applicable taxes.

2. Non-Resident External (NRE) Account:

- **Repatriable Funds:** Both principal and interest can be repatriated without any restrictions.
- **Tax-Free Interest:** Interest earned is not taxed in India.
- **Foreign Earnings:** Only foreign earnings can be deposited.

3. Foreign Currency Non-Resident (FCNR) Account:

- **Foreign Currency:** Maintained in foreign currency (USD, GBP, EUR, etc.).
- **Tax-Free Interest:** Interest earned is not taxed in India.
- **Repatriable Funds:** Both principal and interest are fully repatriable.
- **Ideal For:** Non-resident Indians looking to manage their income in India while enjoying the benefits of repatriation and tax exemptions.

● Type of Accounts

■ Individual Accounts

An individual bank account, like a savings or current account, is a financial account held by a single person, offering features like deposits, withdrawals, and potentially interest earnings.

■ Joint Accounts

A joint account is a bank account shared by two or more individuals, granting each holder equal access and ownership to the funds, commonly used by couples, families, or business partners.

■ Business/Corporate Accounts

A business corporate account is a bank account that a company uses to manage its finances. It's also known as a business account or a company account.